Name: _____ Register No: _____

Experiment 20

Question: Experiment to understand K-means clustering algorithm for a given data set.

Aim

To understand the K-means clustering algorithm and group unlabelled data into clusters.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

def exp_20():
    from sklearn.datasets import make_blobs
    from sklearn.cluster import KMeans
    from sklearn.metrics import silhouette_score

    X, _ = make_blobs(n_samples=300, centers=4, cluster_std=0.9, random_state=7)
    print("Data shape:", X.shape, "\n")

    # Elbow method and silhouette score to choose K
    inertia, sil = [], []
    ks = range(2, 9)
    for k in ks:
        km = KMeans(n_clusters=k, n_init=10, random_state=42).fit(X)
        inertia.append(km.inertia_)
        sil.append(silhouette_score(X, km.labels_))
    print("K   Inertia(WCSS)   Silhouette")
    for k, i, s in zip(ks, inertia, sil):
        print(f"{k}   {i:10.2f}      {s:.4f}")
    best_k = list(ks)[int(np.argmax(sil))]
    print("\nBest K by silhouette score:", best_k)

    kmeans = KMeans(n_clusters=best_k, n_init=10, random_state=42)
    labels = kmeans.fit_predict(X)
    print("\nCluster sizes:", {int(k): int(v) for k, v in zip(*np.unique(labels,
        return_counts=True))})
    print("Cluster centres:\n", kmeans.cluster_centers_.round(3))
    print("Inertia:", round(kmeans.inertia_, 2))
    print("Silhouette score:", round(silhouette_score(X, labels), 4))

    fig, ax = plt.subplots(1, 2, figsize=(12, 4.8))
    ax[0].plot(list(ks), inertia, marker="o")
    ax[0].axvline(best_k, color="red", linestyle="--", label=f"K = {best_k}")
    ax[0].set(title="Elbow Method", xlabel="Number of clusters (K)", ylabel="Inertia (WCSS)")
    ax[0].legend()
    ax[1].scatter(X[:, 0], X[:, 1], c=labels, cmap="viridis", s=25)
    ax[1].scatter(*kmeans.cluster_centers_.T, c="red", marker="X", s=200, edgecolor="black",
        label="Centroids")
    ax[1].set(title=f"K-means Clusters (K = {best_k})", xlabel="Feature 1", ylabel="Feature 2")
    ax[1].legend()
    plt.tight_layout()
    plt.show()


exp_20()
```

Output

```
Data shape: (300, 2)

K   Inertia(WCSS)   Silhouette
2      8941.64      0.5995
3      2402.84      0.7757
4       462.76      0.8265
5       416.09      0.7212
6       369.18      0.5894
7       322.48      0.4962
8       287.12      0.3511

Best K by silhouette score: 4
Cluster sizes: {0: 75, 1: 75, 2: 75, 3: 75}
```

```
Cluster centres:
 [[-1.314  4.544]
 [ 0.088 -8.711]
 [ 9.521  0.53 ]
 [-8.445  5.487]]
Inertia: 462.76
Silhouette score: 0.8265
```

Graph

